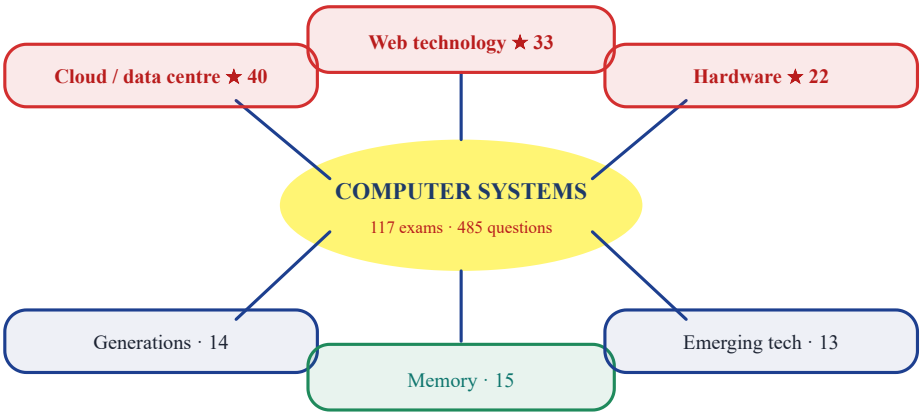
Introduction to Computer Systems

every computer-fundamentals, cloud, web and hardware question asked in BCS · NTRCA · BPSC · bank · power-sector · BUET-set papers, solved in 32 sheets



↑ the number is how many different exams asked that topic (red = most asked)

How to read a question card

> **Q:** The question, as printed in the paper (typos fixed).
>
> ✎ **Asked in:** the exams it was taken from, e.g. BPSC AP 2022 · 45th BCS ICT (2024)

Every HTML / CSS / JS snippet and every Excel formula was run; green = the real output. Numerical answers (VM count, reliability) are worked on the sheet.

Most asked

① System vs application software · 4
② Cloud computing & service models · many
③ VM / hypervisor · 2+ (BB 2023)
④ Data-centre energy efficiency · 2
⑤ HTML element / page structure · CSS box model

Syllabus coverage

"Computer **generations & classification**, hardware & peripherals, memory, **software**, BIOS/booting, **cloud computing** & data centres, virtualization, **web technology** (HTML/CSS/JS, HTTP), e-commerce, emerging technology and ICT in society."

> **Legend:** **yellow** = key term · **red circle** = must-memorise · **purple box** = memory trick · **red box** = exam trap · **green** = checked answer

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1 System software vs application software

Q1: What is the difference between system software and application software? / Explain software with examples. / Difference between hardware and software.

✎ **Asked in:** 45th BCS General (Sci & Tech) (2024) · BB AME 2016 · MM AP 2017 · NPCBL JAM 2022 · 40th BCS (2020) · 15th NTRCA Computer Operation (2019)

Point	System software	Application software
job	runs & manages the **computer itself** (always, in background)	does a **user task** (when opened)
examples	**OS** (Windows, Linux), compiler, driver, utility	**MS Word, Excel**, browser, banking app

Software = instructions that tell the hardware what to do (no physical form); **hardware** = the physical parts. A **utility** (antivirus, backup) is system software; **firmware** = software permanently in ROM (BIOS).

Q2: What is platform-independent software? Discuss with an example.

✎ **Asked in:** Combined Bank SO(IT) 2023

Platform-independent software runs on any OS/hardware unchanged, because it runs on a virtual machine, not the bare CPU. **Java**: source → `bytecode` → the **JVM** on each platform ("write once, run anywhere"). Also Python and any web page.

Q3: What is version control (e.g. Git)? Explain the difference between committing and pushing code.

✎ **Asked in:** Bank (name cut off) AP

Version control (Git) tracks code changes and lets teams roll back. **commit** = save a snapshot to your *local* repo; **push** = send local commits to the *remote* (GitHub). Commit is local, push is network.

Q4: Describe the classification of computer software. / List banking software used in Bangladesh and the features of good banking software.

✎ **Asked in:** Intl Crimes Tribunal Hardware Technician 2024 · Combined Bank Officer(IT) 2020

Software → **system** (OS, translators, utilities) + **application** (Office; custom core-banking, e.g. Flora, Ababil, T24). Good banking software: **security**, 24×7 uptime, real-time, audit trail, scalable, compliant.

Trap: a **compiler/assembler is system software** (serves the machine), not an app; **antivirus = utility = system software**.

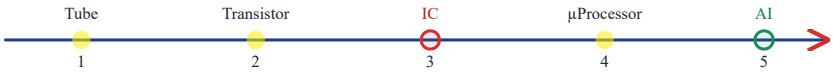
2 Generations of computers (1st–5th)

Q5: What is a computer generation? Describe the characteristics of the different generations. / Give the features of the 2nd, 3rd and 5th generations.

✎ **Asked in:** 18th NTRCA School (2024) · 15th NTRCA Computer Operation (2019) · 13th NTRCA CSE (2016) · 17th NTRCA School (2023)

Gen	Switching device	Era	Features / example
1st	**Vacuum tube**	1946–59	huge, hot, machine language · ENIAC, UNIVAC
2nd	**Transistor**	1959–65	smaller, less heat, assembly/COBOL · IBM 1401
3rd	**IC**	1965–71	multiprogramming, keyboard/monitor · IBM 360
4th	**Microprocessor** (VLSI)	1971–now	PC, GUI, network · Intel 4004, IBM PC
5th	AI / ULSI	now→	parallel, AI, natural language · quantum

Trick: Tube → Transistor → IC → Microprocessor → AI. Each generation = smaller, faster, cheaper, less power.



↑ the switching device defines the generation

MCQ as asked	answer	asked in
Microprocessor was first used in which generation?	**Fourth (4th)**	DESCO AE(CSE) 2016
In which year were chips (ICs) first used inside a computer? 1964 / 1974 / 1975 / 1981	**1964** (3rd gen, IBM 360)	Combined Bank SO(IT) 2021
A webcam is a ___ device (System / Analogue / **Input** / Output)	**Input**	BB AD(General) 2023
The word "computer" means a ___ machine (হিসাবকারী / **গণনাকারী** / পরীক্ষার / বিমান)	**গণনাকারী (calculating)**	18th NTRCA College Prelim (2024)

Memory trick: "Electricity" marker: vacuum tube (1), transistor (2), IC (3), microprocessor (4), AI (5). Any question naming a **transistor** ⇒ 2nd; **IC** ⇒ 3rd; **microprocessor** ⇒ 4th.

3 Analog / digital / hybrid · micro to super

Q6: Classify computers by working principle. / Difference between analog and digital computer. / In 3 fields, give the use of hybrid computers.

✎ **Asked in:** 17th NTRCA School (2023) · 15th NTRCA CSE

	Analog	Digital	Hybrid
works on	continuous physical quantity	discrete **0/1** digits	both
example	speedometer, thermometer	**PC, laptop**	ICU monitor, petrol pump
accuracy	lower	high	high

Hybrid use (3 fields): hospital ICU (measures BP/ECG as analog, shows digital), industrial process control, petrol pump, defence/aircraft.

> **Q7:** Classify computers by size, shape and capacity. / Discuss super computer and mainframe. / Components of a micro-computer system.
> ✎ **Asked in:** 16th NTRCA CSE (2019) · 17th NTRCA School (2023) · BPSC ADA 2022

Type	Size / power	Use
Super	fastest, parallel	weather, nuclear, research (BD: BMD)
Mainframe	large, many users	bank, census, airline booking
Mini	medium	mid-size firm server
Micro (PC)	single user	desktop, laptop, mobile

Micro-computer system = microprocessor (CPU) + memory (RAM/ROM) + I/O devices + system bus + power supply, in one box. **Super vs mainframe:** super = max *speed* on one big job; mainframe = max *throughput* for many users/transactions.

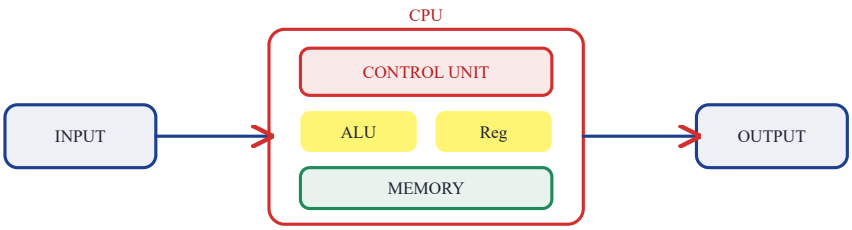
By purpose: general-purpose (PC) vs special-purpose (ATM, washing-machine controller — embedded). **Workstation** = a powerful single-user PC for engineering/graphics. **Server** = a computer that provides resources/services to clients over a network. **Embedded system** = a computer built into a device (car, router, smart TV).

> **Memory trick:** Order by power: **Micro < Mini < Mainframe < Super**. By signal: **Analog (continuous) · Digital (0/1) · Hybrid (both)**.

> **Exam trap:** A **mainframe** serves thousands of users at once; a **supercomputer** throws all its power at one scientific problem. They are not the same "big computer".

4 Building blocks & how a computer works

> **Q8:** What are the building blocks of a digital computer? Describe each unit briefly. / Draw the block diagram and describe each unit. / Explain the steps of a computer's working.
> ✎ **Asked in:** 38th BCS (2018) · 47th BCS CSE · 17th NTRCA CSE (2023) · 15th NTRCA CSE



↑ Input → CPU (CU + ALU + registers) ↔ Memory → Output; secondary storage holds data permanently

Units: Input (keyboard, mouse) feeds data; **CPU** = **Control Unit** (fetch–decode–control) + **ALU** (arithmetic + logic) + **registers**; **Memory** holds the program/data being used; **Output** (monitor, printer) shows results; **secondary storage** keeps data permanently. **Working = IPO(S):** Input → Process → Output → Storage; the CPU runs the **fetch–decode–execute** cycle.

> **Q9:** What are the components of a micro-computer system? Explain the main organisational parts with a diagram.
> ✎ **Asked in:** BPSC ADA 2022 · 17th NTRCA CSE (2023) · 15th NTRCA CSE

CPU (microprocessor) + primary memory (RAM, ROM) + I/O devices + storage + the three **buses** (data, address, control) on the motherboard — see the bus sheet.

> **Q10:** What is the total number of bits in IPv6? What do DIMM, TDP, SSD and HTTP stand for? / What criteria does computer speed depend on?
> ✎ **Asked in:** BUET ICT Cell AHE 2025 · BTCL JAM 2024

IPv6 = 128 bits. DIMM Dual In-line Memory Module · **TDP** Thermal Design Power · **SSD** Solid State Drive · **HTTP** HyperText Transfer Protocol. Computer speed depends on: **clock speed** (GHz), word size/bus width, number of cores, cache, RAM size and storage type (SSD vs HDD).

Fetch–decode–execute cycle: (1) the Control Unit **fetches** the next instruction from memory (address in the PC); (2) **decodes** it in the IR; (3) the ALU **executes** it; (4) the result is written back and the PC advances. This loop, millions/s, is all a computer does.

> **Memory trick:** Remember **IPO-S**: Input → Process (CPU) → Output → Storage. Every "how a computer works" answer is these four units plus the fetch–decode–execute cycle.

5 CPU, GPU, system bus, interrupts

> **Q11:** What is CPU? Describe the organisation of the CPU. / What is the function of a GPU?
> ✎ **Asked in:** 17th NTRCA School (2023) · BPSC AP 2022

CPU ("brain") = **CU** (directs all operations) + **ALU** (arithmetic + logic/compare) + **registers** (fastest tiny stores: PC, IR, AC, MAR, MBR). **GPU** = many small cores for parallel graphics/matrix maths; used for 3D, video and **AI/ML** training. CPU = few fast general cores; GPU = thousands of simple cores.

Q12: What is a computer bus? What is the unit of its speed? Describe the types of bus. / Show how RAM, ROM and I/O connect to a microprocessor by bus. / Components of a USB bus with block diagram.

✎ **Asked in:** BKKB AP 2017 · BPSC SAE 2021 · ICT Ministry AP 2017

Bus	carries	direction
Data bus	the actual data	two-way
Address bus	memory address	one-way (CPU→)
Control bus	read/write, clock, interrupt signals	two-way

A **bus** is a set of parallel wires carrying bits between CPU, memory and I/O. Speed is measured in **MHz/GHz** (bus clock) or bytes/s (bandwidth = width × clock). **USB** (Universal Serial Bus) = serial, hot-pluggable, powers the device; a USB packet = sync + PID + address/endpoint + data + CRC.

Q12b: Write down the difference between a serial port and a parallel port.

✎ **Asked in:** DESCO SAE 2023

Serial port sends 1 bit at a time over one wire (RS-232, USB) — fewer wires, good for long distance; **parallel port** sends many bits at once over many wires (old LPT/printer) — faster over short runs but bulky and prone to skew. Modern systems are nearly all serial (USB, SATA).

Q13: How can an interrupt be enabled? What steps occur when an interrupt is processed?

✎ **Asked in:** 38th BCS (2018)

1. Device raises an **interrupt request (IRQ)** on the control bus (enabled by setting the interrupt-enable flag).
2. CPU finishes the current instruction, saves **PC + registers** on the stack.
3. Jumps to the **ISR** (interrupt service routine) via the interrupt vector.
4. ISR services the device, then **IRET** restores the saved state and resumes.

Q14: Differentiate among Intel Core i3, i5 and i7 processors.

✎ **Asked in:** Combined Bank SO(IT) 2023

Rising cores/threads, cache and Turbo Boost: **i3** entry (2–4 cores, no/low turbo), **i5** mid (4–6 cores, turbo), **i7** high (6–8+ cores, hyper-threading, big cache). More cores + cache + turbo = faster but costlier/hotter.

6 Optical & magnetic input: OCR, OMR, MICR, RFID

Q15: What is OCR? Difference between OCR and OMR. / Distinguish between OMR and MICR. / Name two optical input devices.

✎ **Asked in:** NPCBL JAM 2022 · BB AME 2016 · BPSC ADA 2022

Device	Full form / reads	Use
OCR	Optical Character Recognition — printed/handwritten **characters**	scan text to editable text
OMR	Optical Mark Reader — pencil **marks**	MCQ/OMR answer sheets
MICR	Magnetic Ink Character Recognition — magnetic ink digits	**bank cheques**
Barcode/RFID	light / radio waves	POS, inventory, toll

MICR is at the bottom of a cheque (E-13B font); it survives stamps/signatures because it is read magnetically, not optically — that is why banks use it. **OMR** reads *where* a mark is (not what it says); **OCR** reads the *characters*.

> **Q16:** RFID has huge applications in supply-chain and toll collection. Show the basic working principle of RFID.
> ✎ **Asked in:** Combined Bank Officer(IT) 2020

RFID = Radio-Frequency IDentification: a **tag** (chip + antenna) + a **reader**. The reader sends radio waves; a passive tag draws power from them and transmits its stored ID back; the reader sends it to the system. No line-of-sight needed (unlike barcode). Use: toll (e-toll), library, inventory, access cards.

MCQ as asked	answer	asked in
Which device is used for primary input of OCR?	**Scanner**	BB AD(General) 2018
In MICR, "C" stands for —	**Character**	Teletalk Executive 2019
Which is NOT contained in a MICR code line?	**Signature** (it has bank/branch/account/cheque no.)	Combined Bank Officer(IT) 2020
A bar-code reader emits —	**Light (laser/LED)**	Combined Bank Officer(IT) 2020
Which is true for RFID?	**No line-of-sight; uses radio waves**	Combined Bank Officer(IT) 2020

7 Printers, monitor, resolution

> **Q17:** Explain impact vs non-impact printer. / Describe the working of an inkjet / laser printer with a diagram. / What is a non-impact printer?
> ✎ **Asked in:** Combined Bank SO(AME) 2021 · Combined Bank AHE 2021 · 18th NTRCA School-2 (2024) · 15th NTRCA Computer Operation (2019)

	Impact	Non-impact
prints by	hammer/pins **strike** an inked ribbon	ink/toner, **no contact**
examples	dot-matrix, line, daisy-wheel	**inkjet, laser**, thermal
noise / speed	noisy, slow	quiet, fast
carbon copy?	**yes** (multi-part)	no

Inkjet: tiny nozzles spray charged ink droplets onto paper. **Laser:** a laser draws the image on a charged drum → drum picks up toner → toner transferred to paper → fused by heat. **Dot-matrix (impact)** printers are used at **bank cash counters** because they make carbon copies of vouchers.

> **Q18:** What do you understand by the resolution of a computer screen? / Factors to consider when buying a monitor. / How does an optical mouse work?
> ✎ **Asked in:** BPSC AP 2022 · BB AME 2016 · 36th BCS (2016)

Resolution = number of pixels (width × height, e.g. 1920×1080); more pixels = sharper. Buy a monitor by: size, resolution, panel (IPS/VA), refresh rate, response time, ports, eye comfort. **Optical mouse:** an LED lights the surface, a tiny camera takes thousands of images/s, a DSP computes the movement.

MCQ as asked	answer	asked in
Which printer is used at a bank cash counter / in banks & offices?	**Dot-matrix** (counter) · **Laser** (office)	Combined Bank Officer(IT) 2026
Printing quality is measured in — (ppm / **dpi** / cpi / cps)	**dpi** (dots per inch)	BB AD(General) 2023
Which first when troubleshooting a faulty monitor?	**Check its connections** to PC & power	Combined Bank AHE/AE(IT) 2021
___ helps prevent power surges	**Surge protector / UPS**	Combined Bank SO(IT) 2021

> **Exam trap:** "ppm" is printer *speed* (pages per minute); **dpi** is print *quality*. A camera for bank surveillance is a **CCTV / IP camera**.

8 RAM / ROM, SSD / HDD, optical disk

> **Q19:** Difference between RAM and ROM. / Why is RAM unsuitable as permanent storage?
> ✎ **Asked in:** 15th NTRCA CSE · BTRC AD(Technical) 2019 · BTRC SAD 2021

	RAM	ROM
volatile?	**yes** — lost on power off	no — permanent
read/write	read & write	read only (mostly)
holds	running programs/data	BIOS/firmware
types	SRAM (cache), DRAM (main)	PROM, EPROM, EEPROM

RAM is unsuitable as permanent storage because it is **volatile** — it loses its contents when power is off. Memory hierarchy (fast→slow, small→big): **registers → cache → RAM → SSD/HDD**.

> **Q20:** Classify common storage devices. What is a solid-state device? / How does an optical disk read/write data? / What is MIDI? Describe the structure of a CD-ROM.
> ✎ **Asked in:** 45th BCS ICT (2024) · BPSC ANE 2020 · 15th NTRCA Computer Operation (2019)

	HDD (magnetic)	SSD (solid-state)	Optical (CD/DVD)
stores by	magnetising platters	charge in flash cells	pits burned, read by **laser**
moving parts	yes (arm, platter)	**none**	spins
speed / shock	slower, fragile	**fast, durable**	slow

SSD = flash memory, no moving parts → faster, silent, durable. **Optical** disk: a laser burns/reads pits & lands on a spiral track; CD → DVD → Blu-ray = shorter wavelength, more capacity. **MIDI** = Musical Instrument Digital Interface (stores notes/commands, not audio).

MCQ as asked	answer	asked in
Which is main memory? RAM / ROM / HDD / Floppy	**RAM**	DoICT AP 2020
SSDs are more durable in harsh environments because —	**They have no actuator arms / moving parts**	Combined Bank SO(IT) 2021
EDSAC stored data using what memory?	**Mercury delay lines**	Combined Bank ADA 2021
Best way to protect hard-drive data	**Regular backup**	Combined Bank AHE/AE(IT) 2021

9 Bits, bytes, units; data vs information

> **Q21:** Difference between data and information. / Difference between MB and Mbps. / Discuss data-processing methods.
> ✎ **Asked in:** SPCBL SAE 2022 · WZPDCL SAE 2023 · 17th NTRCA School (2023)

Data = raw, unorganised facts; **information** = processed, meaningful data. **MB** (megabyte) = a *storage* size (8 bits = 1 byte); **Mbps** (megabits per second) = a *speed*. 1 byte = 8 bits, so 100 MB ≠ 100 Mb. Data processing: collection → input → process → output → storage (batch, online, real-time, distributed).

Unit	=	Unit	=
1 byte	8 bits	1 GB	1024 MB
1 KB	1024 B	1 TB	1024 GB
1 MB	1024 KB	nibble	4 bits

MCQ as asked	answer	asked in
A Unicode character uses how many bits? 8/16/20/24	**16**	BCC AP 2021
1 TB equals how many bytes?	**10^{12} (SI)**; 2^{40} = 1,099,511,627,776 (binary)	BB AD(General) 2022
Exact bytes in 960 Kb main memory	**960 × 1024** bytes	BPSC AP 2022
1 megabyte = how many kilobytes? 1000/512/1026/1024	**1024**	BPSC AME 2017
Correct order by size	**byte < KB < MB < GB < TB**	BDCCL AM(Transmission) 2022
Unit of hard-disk capacity (MHz/kHz/**Gigabyte**/None)	**Gigabyte**	BCC AP 2021
Typical speed of USB 3.0	**4.8 Gbit/s** (≈ 5 Gbps)	Combined Bank SO(IT) 2021
GPS stands for	**Global Positioning System**	BCC AP 2021

Number systems (covered in the Number Systems lesson): binary (base 2), octal (8), decimal (10), hex (16). Character codes: **ASCII** = 7/8-bit (English); **Unicode (UTF-8/16)** covers all scripts incl. Bangla; **EBCDIC** is the old IBM 8-bit code. Full units chain: bit → nibble (4) → byte (8) → KB → MB → GB → TB → PB → EB.

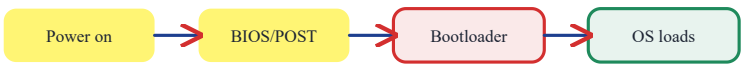
Exam trap: In computing, 1 KB = **1024** bytes (binary), but disk makers use SI 1 KB = 1000 B — which is why a "1 TB" drive shows ~931 GB. Write both forms if the options allow.

10 BIOS, booting, POST, CMOS

Q22: What is BIOS? What is its main role in a computer? / Explain BIOS in a server and how it affects booting in hardware maintenance.

✎ **Asked in:** EGCB SAE(ICT) 2023 · 45th BCS General (Sci & Tech) (2024) · RAKUB ANSE 2023

BIOS = Basic Input/Output System — **firmware stored in ROM** on the motherboard. On power-on it: (1) runs **POST** (Power-On Self-Test) to check RAM, keyboard, disk; (2) initialises hardware; (3) finds the boot device and loads the **bootloader** → the OS. Settings (boot order, date, passwords) are stored in **CMOS**, kept alive by a small battery. On servers, BIOS/UEFI sets RAID, boot order and virtualization flags — wrong settings stop the OS loading.



↑ the boot sequence: a single beep = POST passed

MCQ as asked	answer	asked in
BIOS stands for —	**Basic Input / Output System**	BTCL JAM 2022
Which is a WRONG statement about BIOS? (connects CPU & I/O / manages data flow / loads OS / **provides storage**)	**"Provides storage"** (it does not)	Combined Bank AHE/AE(IT) 2021
No beep on boot — what to check first?	**Power supply**	Combined Bank AHE/AE(IT) 2021
How to clear a CMOS password?	**Remove the BIOS battery / change the jumper**	Combined Bank AHE/AE(IT) 2021

BIOS vs UEFI: UEFI is the modern replacement — supports disks > 2 TB (GPT), a graphical setup, faster boot and Secure Boot. **Cold boot** = starting from power-off; **warm boot** = restarting without cutting power (Ctrl+Alt+Del). **Bootstrapping** = the small program that "pulls the system up by its bootstraps" to load the OS.

> **Memory trick:** Order on power-on: **Power → POST → BIOS → Bootloader → OS**. One short beep = all hardware OK; repeated beeps = a hardware fault (RAM/GPU) by beep code.

11 Cloud computing & IaaS / PaaS / SaaS

> **Q23:** What is cloud computing? Write its advantages and disadvantages / service models. / How does it differ from traditional computing? / Write the three basic functions of cloud services.
> ✎ **Asked in:** 17th NTRCA CSE (2023) · 17th NTRCA ICT (2023) · NPCBL JAM 2022 · BPSC AP 2021 · Parliament Secretariat SAE 2023 · NPCBL Engineer(Cyber Security) 2026 · BOF AE(CSE) 2021 · BLPU AP 2018

Three basic functions / essential characteristics of cloud services (NIST): **on-demand self-service**, **broad network access** and **resource pooling** (plus rapid elasticity and measured/pay-per-use service).

Cloud computing = delivering computing resources (servers, storage, databases, software) **over the internet, on demand, pay-as-you-go**, instead of owning hardware. **Advantages:** no upfront cost, elastic scaling, anywhere access, automatic backup/updates, high availability. **Disadvantages:** needs internet, data-security/privacy concerns, vendor lock-in, recurring cost, less control. **Deployment:** public, private, hybrid, community.

Model	You get	You manage	Example
IaaS	VMs, storage, network	OS, runtime, app	AWS EC2, Azure VM
PaaS	+ OS & runtime platform	just your app/code	**Heroku, App Engine**
SaaS	ready software	nothing (just use)	Gmail, Office 365

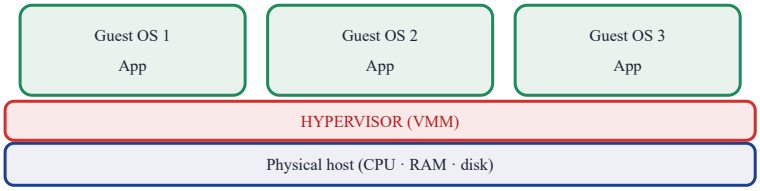
Trap scenario (Combined Bank SO(IT) 2026): a startup wants to deploy code only, not manage OS/runtime/hardware → **PaaS** (e.g. Heroku, Google App Engine). **5 enabling technologies of cloud:** virtualization, broadband network, distributed/grid computing, service-oriented architecture, web services/APIs, automation. Pricing/deploy often: **IaaS** = rent infrastructure; **SaaS** = rent software.

MCQ as asked	answer	asked in
Which is NOT a layer of cloud computing? (**CaaS** / IaaS / PaaS / SaaS)	**Computing as a service (CaaS)**	Combined Bank AHE/AE(IT) 2021
Which will NOT be treated as a cloud service?	**A web portal self-hosted by PSC**	BPSC AME 2022
Not a reason to use colocation for DR instead of building a DC	**Full control of hardware**	Combined Bank ADA 2021

12 Virtual machine, hypervisor, container

Q24: Define a virtual machine with a diagram, explain how a VM works, and list its benefits. / What is a hypervisor and its types? / Define virtualization; role of hypervisor; VM vs container.

✎ **Asked in:** BB AME 2023 · BB AP 2023 · BUET ICT Cell AHE 2025 · Combined Bank AP(SB & Janata) 2026 · RAKUB ANSE 2023 · Parliament Secretariat SAE 2023



↑ one physical host runs several isolated VMs, each with its own guest OS, over a hypervisor

VM = a software computer with its own virtual CPU/RAM/disk and **its own guest OS**, running on a **hypervisor (VMM)** that shares the real hardware. **Benefits:** isolation, run many OSes on one box, easy snapshot/backup/recovery, fast provisioning, better utilisation. **Hypervisor types: Type 1 (bare-metal)** runs directly on hardware (ESXi, Hyper-V, Xen) — fast, for servers; **Type 2 (hosted)** runs on a host OS (VirtualBox, VMware Workstation) — for desktops.

	Virtual machine	Container (Docker)
includes	a full **guest OS**	app + libs only, **shares host OS kernel**
size / start	GBs, boots in minutes	**MBs, starts in seconds**
isolation	strong (own OS)	lighter (process-level)

Docker = container platform; **Docker Hub** = cloud image repository. **Kubernetes (K8s)** orchestrates containers (control plane + worker nodes/pods). **JVM** runs Java bytecode portably — it is not a hypervisor.

Q25b: For reliable, efficient data transmission, VMs or containers? / Which architecture for a system monitoring a third-party cloud service through APIs?

✎ **Asked in:** BSCL AE(CSE) 2026

VMs vs containers: prefer **containers** — lower overhead, faster start, easy scaling and CI/CD; use VMs only for strong OS-level isolation or a different guest OS. **API monitoring:** an **event-driven microservices** design (services subscribing to the APIs + a queue + a dashboard) scales and isolates faults.

Q25: A physical server has 32 CPU cores, 96 GB RAM, 4 TB storage. Each VM needs 4 cores, 16 GB RAM, 500 GB. Max VMs without over-committing? Which resource is the limit?

✎ **Asked in:** Combined Bank Officer(IT) 2026

CPU: 32/4 = **8** · RAM: 96/16 = **6** · Storage: 4000/500 = **8**. Max = min(8, 6, 8) = **6 VMs**; **RAM is the limiting resource.**

13 Data centre, energy, maintenance, DR

Q26: What are the challenges in optimising the energy efficiency of a data centre? / Explain the components of a data centre. / Key hardware components of a typical server.

✎ **Asked in:** BB AME 2023 · BB AP 2023 · ABL SO(IT) 2017 · Rupali Bank ANA 2023

Data-centre components: servers (compute), storage/SAN, network (switches, routers, firewall), **power** (UPS, generator, PDU), **cooling** (CRAC/HVAC), racks, fire suppression, physical security. **Energy-efficiency challenges:** cooling is the biggest draw, low average server utilisation, power-distribution losses, legacy/inefficient equipment, measuring and managing **PUE** (Power Usage Effectiveness = total power ÷ IT power; ideal → 1.0). Fixes: virtualization/consolidation, hot/cold-aisle containment, free cooling, efficient PSUs.

Q27: Distinguish preventive vs corrective maintenance in a data centre (with an example). / What should be checked before buying servers / a data centre? / DR & business-continuity plan for your data centre.

✎ **Asked in:** Combined Bank AME 2026 · Combined Bank ADA 2026 · Combined Bank ADA 2024 · Combined Bank AME/AHE 2023 · Combined Bank SO(IT) 2024

	Preventive	Corrective
when	**before** failure, scheduled	**after** a failure
example	cleaning, patching, replacing aging UPS battery	swap a dead disk, restart a crashed service
goal	avoid downtime	restore service fast

Before buying servers/a DC: workload/capacity, CPU/RAM/storage, redundancy (power, cooling, network), scalability, Tier level, warranty/support, security, cost/TCO, location. **DR/BCP:** define **RTO** (how fast to recover) & **RPO** (how much data loss is acceptable), backups (3-2-1 rule), a hot/warm/cold standby site, regular drills. **Reliability** of 3 series components each 0.95 = 0.95^3 = **0.857**.

Q28: What is the need for an edge server? / What do you mean by a National Data Center?

✎ **Asked in:** Petrobangla AM(CSE) 2024 · BPSC AP 2022

Edge server processes data near the source (not a far cloud): lower **latency**, less bandwidth, real-time response, local privacy (CDN, IoT, CCTV). **National Data Center** (BD: Kaliakoir, Tier-4) hosts government data/services centrally with high security and uptime.

14 Scaling, web-app deployment, big data

Q29: Difference between vertical and horizontal scaling. What is auto-scaling and how does it work? / Draw a server rack showing horizontal and vertical scaling.

✎ **Asked in:** BUET ICT Cell AHE 2025 · RPGCL AM(ICT) 2022

	Vertical (scale up)	Horizontal (scale out)
means	bigger CPU/RAM in the **same** server	**more servers/nodes** behind a load balancer
limit	one machine's max	near-unlimited
downtime	usually needs restart	none (add nodes live)

Auto-scaling watches load (CPU/requests) and automatically adds nodes when busy and removes them when idle, so you pay only for what you use and stay available at peak.

Q30: Describe a typical web-application deployment architecture and how the tiers interact. / Sketch a plan to transform a data centre into a cloud. / What is an API Gateway and how does it interact with backend microservices?

✎ **Asked in:** Combined Bank AME/AHE 2023 · Combined Bank AME/AE(IT) 2024 · Combined Bank ADA 2026

3-tier web app: client (browser) → **web/presentation tier** (load balancer + web servers) → **application tier** (business logic, APIs) → **data tier** (database). An **API Gateway** is the single entry point in front of microservices: it routes requests, authenticates, rate-limits, aggregates responses and hides the internal services. **DC→cloud plan:** assess apps → choose model (IaaS/PaaS) → pick strategy (rehost/replatform/refactor) → migrate data → test → cut over → optimise.

Q31: Write ten sentences on "Big Data". What is Big Data and its characteristics? / Identify 5 Hadoop-ecosystem components for processing massive server logs. / SaaS & multi-tenant architecture: advantages and disadvantages.

✎ **Asked in:** MRA SCO 2023 · BPSC AP 2021 · Combined Bank AME 2026 · Combined Bank SO(IT) 2024

Big Data = data sets too large/fast/varied for ordinary tools; the **5 V's**: Volume, Velocity, Variety, Veracity, Value. **Hadoop ecosystem: HDFS** (storage), **MapReduce/YARN** (processing/resource mgmt), **Hive** (SQL on big data), **HBase** (NoSQL), **Spark** (fast in-memory), **Kafka** (log/stream ingestion). **Multi-tenant SaaS** = one app instance serves many customers (tenants); pro = low cost, easy updates; con = weaker isolation, "noisy neighbour", harder per-tenant customisation.

15 Cloud security, threats, server scenarios

Q32: How do security assessments and audit reports detect vulnerabilities and ensure cloud-security-posture compliance? / Cyber-security threats to a microfinance app hosted in your DC, and how to mitigate them. / Tools/technology to secure an e-banking web server in your DC.

✎ **Asked in:** BSCL AE(CSE) 2025 · Combined Bank SO(IT) 2024 · Combined Bank AME/AE(IT) 2024

Cloud-security posture: regular vulnerability scans + penetration tests + audit logs + config review (CSPM) find mis-configurations, open ports, weak IAM, unpatched software, and check compliance (ISO 27001, PCI-DSS). **Threats** to a hosted app: malware/ransomware, SQL injection & **XSS**, DDoS, data breach, insider misuse, phishing. **Mitigation:** firewall + WAF, HTTPS/TLS, strong IAM & MFA, patching, input validation, encryption at rest/in transit, backups, IDS/IPS, least privilege, monitoring.

Q33: Server maintenance best practices (cleaning, monitoring, patching) and how they help longevity. / How do you handle server-related problems for your company? / Explain BIOS in a server and its effect on boot config.

✎ **Asked in:** Rupali Bank ANA 2023 · Rupali Bank ANE 2023 · Combined Bank Officer(IT) 2023 · RAKUB ANSE 2023

Server best practices: keep firmware/OS patched, monitor (CPU/RAM/disk/temperature, alerts), clean dust & check cooling, test backups & failover, review logs & security, document changes, redundant power/RAID. These extend hardware life and keep uptime high.

Q33b: A company's server traffic rises from 10,000 to 50,000 hits but it does not move to the cloud. Discuss the business and technical consequences.

✎ **Asked in:** ICB Securities AP 2026

Technical: on a fixed on-prem server, 5× load causes CPU/RAM saturation, slow response, timeouts, downtime at peak, no elasticity, and costly over-provisioning for peaks. **Business:** poor user experience, lost transactions/revenue, reputation damage, higher capital & maintenance cost, and slow time-to-scale vs a cloud auto-scaling competitor.

Scenario / MCQ as asked	answer	asked in
HTTP status code 500 indicates —	**Internal Server Error**	BB AD(ICT) 2025
Server upload rejected (unsupported file format): which HTTP code?	**415 Unsupported Media Type** (or 400)	Combined Bank ADA 2026
Cloud outage: cloud-provider vs bank responsibility	**Shared responsibility**: provider = infra uptime/SLA; bank = data, access, its own DR	Combined Bank ADA 2026
What is Docker & Docker Hub?	**Docker = container platform; Hub = cloud image repo**	BB AD(ICT) 2025

16 HTML elements, page structure, HTML5 tags

Q34: What is an HTML element? Explain with examples. / Explain the structure of an HTML web page with an example.

✎ **Asked in:** 17th NTRCA CSE (2023) · 17th NTRCA ICT (2023)

An **HTML element** = a start tag + content + end tag, e.g. `<p>Hello</p>`. Some are empty (`<br>`, `<img>`). A page's skeleton:

```
<!DOCTYPE html>
<html>
 <head><title>My Page</title></head>
 <body>
  <h1>My First Heading</h1>
  <p>My first paragraph.</p>
 </body>
</html>
```

Renders: a big bold "My First Heading" then the line "My first paragraph." (a WZPDCL SAE 2023 trace: a red paragraph via `style="color:red"` shows in red).

Q35: Describe the HTML5 tags <header> <footer> <section> <article>. / Name HTML5 media tags. / How do you change an HTML element's value using the DOM?

✎ **Asked in:** Combined Bank AME 2019 · CB AP 2018

Tag	meaning	Tag	meaning
<header>	top/intro, logo, nav	<section>	a thematic group
<footer>	bottom, copyright	<article>	self-contained content

HTML5 media tags: `<audio>`, `<video>`, `<source>`, `<track>`, `<embed>`. **Change a value via DOM:** `document.getElementById("demo").innerHTML = "New";` (`localStorage` keeps data with no expiry; `sessionStorage` clears when the tab closes — ABL Officer(ICT) 2017). **<canvas>** draws pixels via JS (good for games); **SVG** is vector XML (scales, each shape is in the DOM).

Short Q / MCQ as asked	answer	asked in
Which tag creates a button in HTML?	**<button>** (or <input type="button">)	BCC AP 2023
Which tag is used to write JavaScript in HTML?	**<script>**	BCC AP 2023
The newest version of HTML is —	**HTML5**	Combined Bank Officer(IT) 2020

17 Tables, lists, forms (GET/POST), web structure

Q36: Write HTML for a page with a hyperlink, an ordered list, an unordered list and a form (text box, check box, option, button). / Write the HTML for the given nested list.

✎ **Asked in:** 15th NTRCA ICT · 17th NTRCA ICT (2023) · 17th NTRCA CSE (2023)

```
<a href="page.html">Go</a>
<ol><li>Fruits<ol type="a"><li>Mango</li><li>Orange</li></ol></li></ol>
<ul><li>Vegetables<ul><li>Green capsicum</li></ul></li></ul>
<form method="post">
 <input type="text"> <input type="checkbox">
 <input type="radio" name="g"> <button>Submit</button>
</form>
```

Renders: a link, a numbered list (1. Fruits → a. Mango, b. Orange), a bullet list, and a form row with a text field, a checkbox, a radio button and a Submit button.

Q37: Write HTML for the given table (with rowspan/colspan). / Make a table structure. / What is a website? Describe linear-sequence and tree web structures.

✎ **Asked in:** 18th NTRCA College ICT (2024) · 16th NTRCA ICT · DUET AP 2020

```
<table border="1">
 <tr><th>Name</th><th>Product</th></tr>
 <tr><td rowspan="3">Mr. John</td><td>HP</td></tr>
 <tr><td>ASUS</td></tr><tr><td>Dell</td></tr>
</table>
```

rowspan/colspan merge cells down/across. **Web structures:** *linear* = pages in a fixed sequence (slideshow, sign-up steps); *tree/hierarchical* = a home page branching to sections (most sites).

Q38: Difference between GET and POST; when to use each (with example). / What is web hosting / a web portal? Why is hosting needed?

✎ **Asked in:** 15th NTRCA ICT · BAERA Engineer(CSE) January 2021 · 16th NTRCA ICT

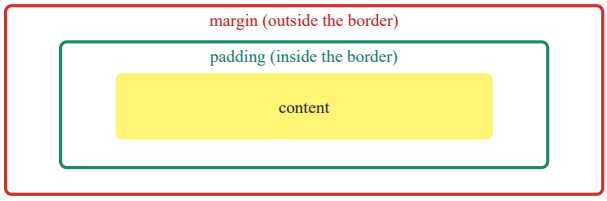
	GET	POST
data in	URL (query string, visible)	request body (hidden)
size / cache	limited, cacheable, bookmarkable	large, not cached
use for	search/read (idempotent)	**login, submit, upload**

Web hosting = renting server space to store a site so it is reachable 24×7 on the internet; needed because a site must live on an always-on, public server. A **web portal** is a gateway page aggregating many services (e.g. a government portal).

18 CSS: box model, types, frameworks, responsive

Q39: In the CSS box model, difference between the Padding and Margin areas. / What is CSS? What is a CSS framework? Name three. / What are the types of CSS?

✎ **Asked in:** 15th NTRCA ICT · 17th NTRCA ICT (2023) · BEPZA Programmer 2023 · 18th NTRCA College ICT (2024)



↑ content → padding → border → margin

Padding = space *inside* the border, between content and border (part of the element's background). **Margin** = space *outside* the border, separating this element from others (transparent). **CSS** (Cascading Style Sheets) styles HTML. **Three types:** inline (`style=""`), internal (`<style>` in <head>), external (`.css` file — best). **Frameworks:** Bootstrap, Tailwind, Foundation, Bulma.

Q40: Explain the role of CSS media queries and the viewport meta tag in responsive apps. What is "mobile-first"? / Why use CSS in a website?

✎ **Asked in:** Combined Bank AME/AE(IT) 2026(Jan) · NESCO AM(ICT) 2021 · ICT Ministry AP 2021

```
<meta name="viewport" content="width=device-width, initial-scale=1">
@media (max-width: 600px) { .menu { flex-direction: column; } }
```

The **viewport meta tag** makes the page use the device's real width (no zoomed-out desktop view on phones). **Media queries** apply different CSS at different screen widths. **Mobile-first** = design the small-screen layout first, then add rules for larger screens with `min-width` queries. **Why CSS:** separates style from content, consistent look, one change updates every page, faster loading, responsive design.

Exam trap: Padding counts as part of the clickable/background box; margin does not. With `box-sizing: border-box`, width includes padding + border (not margin).

19 JavaScript, PHP, web-input security

Q41: Write JS with if-else so that for x = 1..9 it prints "x is an odd value." / "x is an even value." / Difference between == and === in JavaScript.

✎ **Asked in:** PGCB SAE 2021 · BAERA Engineer(CSE) January 2021

```
for (let x = 1; x <= 9; x++) {
  if (x % 2 === 0) console.log(x + " is an even value.");
```

```
  else console.log(x + " is an odd value.");
}
```

1 is an odd value. · 2 is an even value. · … · 9 is an odd value. (odd for 1,3,5,7,9; even for 2,4,6,8)

== compares values with *type coercion* (`5 == "5"` → true); === compares value *and* type, no coercion (`5 === "5"` → false). Always prefer ===.

> **Q42:** What is a closure in JavaScript? Explain with an example. / Write a JS function to validate a 10-digit phone number (digits only, no spaces/+).
> ✎ **Asked in:** ICB AP 2017 · NESCO AM(ICT) 2021

```
function phone(s){ return /^\d{10}$/.test(s.replace(/\D/g,"")); }
console.log(phone("01712345678")); // 11 digits -> false
console.log(phone("1712345678"));  // 10 digits -> true
```

false, true. A **closure** = an inner function that remembers variables from its outer function even after the outer returns (used for private state, counters).

> **Q43:** A CMS blog body inserts un-escaped HTML as `{{body}}`, so a <script> can be injected. Explain the risk and how to prevent it (XSS), with CSP enabled.
> ✎ **Asked in:** DPDC AM(ICT) 2025

Risk = stored XSS (Cross-Site Scripting): if user content like `{{body}}` is placed into the page *without escaping*, a malicious post can run attacker JavaScript in every visitor's browser (steal cookies/sessions, deface, act as the user). **Prevention** (the real exam answer): (1) **output-encode / sanitise** all user content — escape `< > & " '` or use a sanitiser (DOMPurify); never inject raw HTML; (2) a strict **Content-Security-Policy** that blocks inline and third-party scripts; (3) set cookies **HttpOnly** + Secure + SameSite so script cannot read them; (4) validate input and use framework auto-escaping. *(We describe the defence, not a working attack payload.)*

Short Q as asked	answer	asked in
What is PHP? Why is JavaScript needed in web development?	**PHP** = server-side scripting (dynamic pages, DB); **JS** = client-side interactivity/validation	ICT Ministry AP 2021

20 HTTP status codes, SOAP/REST, protocols

> **Q44:** What do the HTTP status-code categories 2xx, 4xx, 5xx represent? Give an example of each. / What does status code 500 indicate?
> ✎ **Asked in:** Combined Bank AME/AE(IT) 2026(Jan) · Combined Bank AME 2026 · BB AD(ICT) 2025

Class	meaning	example
1xx	informational	100 Continue
2xx	success	200 OK, 201 Created
3xx	redirect	301 Moved, 304 Not Modified
4xx	client error	404 Not Found, 401, **415**
5xx	server error	**500 Internal Server Error**

> **Q45:** What are SOAP and RESTful APIs? State one main difference in how they exchange data. / What is SOAP? / Three differences between POST and GET.
> ✎ **Asked in:** Combined Bank Officer(IT) 2025 · BB AP 2019 · BAERA Engineer(CSE) January 2021

	SOAP	REST
data format	**XML only**, strict	JSON/XML/text, flexible
style	protocol, has WSDL	architectural style over HTTP
weight	heavy	**lightweight, fast**

Main difference: **SOAP always exchanges XML messages** (with an envelope); **REST** uses plain HTTP verbs (GET/POST/PUT/DELETE) and usually lightweight JSON.

> **Q46:** Browsing www.psc.gov.bd — list the protocols and their layers used to fetch the page, and the address types at Transport, Network and Data-Link layers.
> ✎ **Asked in:** Parliament Secretariat SAE 2023

DNS (application) resolves the name → IP; **HTTP/HTTPS** (application) requests the page; **TCP** (transport, uses **port numbers**); **IP** (network, uses **IP addresses**); **Ethernet/Wi-Fi** (data-link, uses **MAC addresses**). HTTPS adds TLS for encryption.

MCQ as asked	answer	asked in
Which is NOT a web server? (Apache Tomcat / **PHP** / Jetty)	**PHP** (a language, not a server)	BB AD(ICT) 2025
URL stands for —	**Uniform Resource Locator**	BB AD(General) 2018

21 Going live, deployment, static vs dynamic

> **Q47:** Once a website is designed & developed, what are the steps to make it live online? / Create a webpage and deploy it to an Apache server. / Difference between static and dynamic websites.
> ✎ **Asked in:** Teletalk Executive 2019 · BUET ICT Cell AHE 2025 · 17th NTRCA ICT (2023)

1. Buy a **domain name** and a **hosting** plan (or a cloud VM).
2. Build the site; put files in the web root (Apache: `/var/www/html`).
3. Point the domain's **DNS** (A record) to the server IP.
4. Configure the web server (Apache/Nginx), enable **HTTPS** (SSL cert).

5 Upload via FTP/Git, test, then it is live; monitor & back up.

	Static site	Dynamic site
content	fixed HTML, same for all	generated per request (DB, user)
tech	HTML/CSS/JS	+ server (PHP, Node) + DB
example	a brochure page	**Facebook, a bank portal**

> **Q48:** True/false on HTTP: one page with text + 3 images needs how many request/response messages? Can two sites share one persistent connection? Can one TCP segment carry two HTTP messages? Do responses ever have an empty body?
> ✎ **Asked in:** Combined Bank SO 2020

(i) **True** — 1 request + 1 response for the HTML, then 1 request + 1 response per image = 4 requests, 4 responses. (ii) **False** — a persistent connection is to one host; different hosts need different connections. (iii) **False** — HTTP messages are not split that way across one segment by design. (iv) **False** — responses like 204/304 have an empty body.

Short Q as asked	answer	asked in
What makes it easier to find a visited web page later?	**History / Bookmarks**	BB AD(General) 2018
The first page of a website (in the address bar)	**Home page**	BB AD(General) 2018
Importance of an application framework in web development	**Reusable structure**: faster, secure, maintainable code	BB AME 2019

> **Q48b:** Create 3 tables in an Oracle DB; build one web page with 3 buttons — "Show students" (list from a table), "Show subject-wise marks" (join subject + marks), "Show each student's total" (join all three tables).
> ✎ **Asked in:** BUET AP 2025

Pattern: three tables (`student`, `subject`, `marks`); a page with 3 buttons, each calling a server endpoint that runs one SQL query and renders the result table. Button 2 **joins** subject + marks on `subject_id`; button 3 joins all three and does `GROUP BY student` with `SUM(mark)` for each student's total. (The SQL join detail is drilled in the DBMS set.)

22 Blockchain, nanotech, IR 4.0, quantum

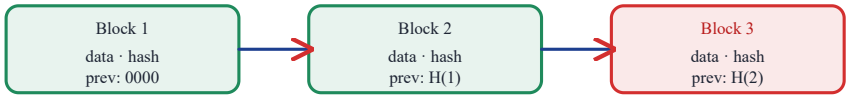
> **Q49:** Write about blockchain: how it works, benefits, uses. / What is nanotechnology, its uses? / What is IR 4.0 and its key components?
> ✎ **Asked in:** WZPGCL AE(CSE) 2023 · Combined Bank SO(IT) 2023 · 17th NTRCA CSE (2023) · ICT Ministry AP 2021 · Fisheries & Livestock Instructor(CS) 2023

Blockchain = a distributed, append-only ledger of blocks, each holding data + a timestamp + the previous block's **hash**, copied across many nodes. Changing one block breaks every later hash, so it is **tamper-evident**; consensus

(PoW/PoS) adds new blocks. **Benefits:** decentralised, transparent, immutable, secure, no middleman. **Uses:** cryptocurrency, smart contracts, supply chain, land records, e-voting.

Nanotechnology = engineering matter at 1–100 nm (atomic/molecular scale). Uses: medicine (targeted drug delivery), electronics (smaller chips), materials (stronger/lighter), energy, water filtration. **IR 4.0** (Fourth Industrial Revolution) fuses the physical, digital and biological worlds; **key components:** AI, IoT, Big Data, cloud, robotics/automation, cyber-physical systems, 3D printing, AR/VR. Unlike IR 1–3 (steam, electricity, computers), IR 4.0's change is **exponential and interconnected**. **Quantum computer** uses qubits (superposition + entanglement) → massively parallel, vs a normal computer's definite 0/1 bits.

Emerging tech	one-line definition
IoT	everyday devices with sensors connected to the internet, sharing data
AI / ML	machines that learn from data to make decisions
AR / VR	overlay digital on real (AR) / fully simulated world (VR)
3D printing	additive manufacturing, layer by layer from a digital model



↑ each block stores the previous block's hash — change one block and every later hash breaks

> **Memory trick:** Each block = **data + own hash + previous hash**. The "previous hash" chain is what makes a blockchain tamper-evident — remember the chain, not the coins.

23 E-commerce, WWW vs internet, search engines

> **Q50:** What is e-commerce? Its types? / Classify e-commerce by nature of transaction. / Difference between e-commerce and m-commerce. / Barriers to e-commerce.
> ✎ **Asked in:** NESCO AM(ICT) 2018 · BPSC ANE 2020 · 17th NTRCA CSE (2023) · 37th BCS · Combined Bank AHE 2021

Type	meaning / example	Type	meaning / example
B2C	business→consumer (Daraz)	C2C	consumer→consumer (Bikroy)
B2B	business→business (Alibaba)	G2C	govt→citizen (e-gov)

E-commerce = buying/selling goods & services online. **M-commerce** = e-commerce on a mobile device (apps, mobile wallet, location-based). **Barriers:** trust/security, low card/internet penetration, logistics/delivery, legal framework, digital literacy, payment-gateway limits.

Q51: What is the WWW and how does it differ from the internet? / E-commerce vs e-business; e-governance.

✎ **Asked in:** Parliament Secretariat SAE 2023 · 15th NTRCA CSE

The **internet** is the global network of networks (the hardware + TCP/IP). The **WWW** is one service running on it — interlinked web pages accessed by HTTP via browsers. Internet also carries email, FTP, etc. **E-business** is wider than e-commerce (all online business processes, not just buying/selling). **E-governance** = delivering government services electronically (G2C, G2B, G2G).

MCQ as asked	answer	asked in
Bangladesh's first internet search engine	**পিপীলিকা (Pipilika)**	16th NTRCA College Prelim (2019)
Which is the first search engine?	**Archie** (1990)	Combined Bank Officer(IT) 2020
Which is NOT a search engine?	**Facebook** (it is social media)	BB AD(General) 2021

Search engine parts: a **crawler/spider** fetches pages, an **indexer** builds a searchable index, and a **query/ranking** engine returns results (Google uses PageRank). Examples: Google, Bing, Yahoo, DuckDuckGo; BD: Pipilika (first Bangla search engine, by SUST, 2013). First-ever: Archie (1990).

Exam trap: Facebook, YouTube and Gmail are **not search engines**. "Browser" (Chrome) ≠ "search engine" (Google): the browser is the app, the search engine is the service you open in it.

24 E-governance, Smart Bangladesh, MIS

Q52: Write short notes: e-Governance; Smart Bangladesh. / Challenges to building Digital Bangladesh. / Challenges to applying MIS in Bangladesh. / Modern-life applications of the internet.

✎ **Asked in:** Parliament Secretariat SAE 2023 · Intl Crimes Tribunal Hardware Technician 2024 · 17th NTRCA School (2023)

E-Governance = using ICT to deliver government services and improve transparency/efficiency; four types: **G2C** (citizen), **G2B** (business), **G2G** (govt–govt), **G2E** (employee). **Smart Bangladesh** (Vision 2041) builds on Digital Bangladesh with four pillars: Smart Citizen, Smart Economy, Smart Government, Smart Society. **Digital Bangladesh challenges:** infrastructure & electricity, affordability, digital literacy, cyber-security, local content, the digital divide.

MIS (Management Information System) turns data into reports for managers' decisions. **MIS challenges in BD:** poor data quality, lack of skilled staff, resistance to change, weak infrastructure, high cost, no integration between systems, data-security gaps. **Internet in modern life:** education, e-commerce, banking (MFS), health (telemedicine), government services, communication, remote work/freelancing, entertainment.

Key BD ICT landmarks: a2i (Access to Information), **e-GP** (electronic government procurement), Union Digital Centres, the National Data Center (Kaliakoir, Tier-4), the ICT Act 2006 / Digital Security Act, and the "Digital

Bangladesh" (2008) → "Smart Bangladesh" (Vision 2041) programme. **Fourth Industrial Revolution vs IR 1–3:** IR1 steam, IR2 electricity/mass production, IR3 computers/automation, IR4 fuses physical–digital–biological systems (AI, IoT, cyber-physical) — change is exponential and networked.

e-Governance type	between	BD example
G2C	govt → citizen	e-passport, land records, MyGov
G2B	govt → business	e-GP (e-tender), VAT online
G2G	govt → govt	inter-ministry file (e-Nothi)
G2E	govt → employee	iBAS++ payroll

> **Model-answer frame:** For any "short note / describe" ICT-society question: **definition → 3–4 features/pillars → BD examples (a2i, MFS, e-GP, Union Digital Centre) → one line on challenges.**

25 Digital banking, electronic payment systems

> **Q53:** Difference between digital and traditional banking. How can digital banking promote financial inclusion? / Describe the transformative power of ICT for online banking (ten points).
> ✎ **Asked in:** Combined Bank SO(IT) 2023 · Combined Bank AME/AHE 2023 · Combined Bank AME/AE(IT) 2024

	Traditional banking	Digital banking
channel	branch, in person	**app / web, 24×7**
reach	limited by branches	anywhere with a phone
cost / speed	higher, slower	lower, instant

Financial inclusion: MFS (bKash, Nagad), agent banking and mobile accounts bring unbanked/rural people into the system — low cost, no branch needed, instant remittance, small savings/loans, government disbursements (allowances, subsidies) direct to a phone.

> **Q54:** Define an electronic payment system (EPS) with a diagram; name five types. / Difference between traditional and electronic payment systems; types of EPS.
> ✎ **Asked in:** BPSC AP 2023 · Intl Crimes Tribunal Hardware Technician 2024

EPS = paying for goods/services electronically, no cash. **Flow:** customer → merchant → **payment gateway** → acquiring bank → card network → issuing bank → back. **Five types:** card (debit/credit), **mobile financial services** (bKash/Nagad), internet/online banking, e-wallet, e-cash/crypto; also NFC/QR. **Traditional vs electronic:** cash/cheque, physical, slow vs digital, instant, trackable, needs internet & security.

Term	meaning (bank IT)
Payment gateway	software that authorises & routes a card/online payment between merchant and banks
NPSB / RTGS / BEFTN	BD inter-bank rails: card switch · real-time large-value · batch EFT
MFS	Mobile Financial Services (bKash, Nagad, Rocket)
KYC	Know Your Customer — identity verification before an account

> **Model-answer frame:** EPS essay: definition → the gateway flow (6 actors) → 5 types with a BD example each → security (encryption, OTP, 2FA, PCI-DSS) → one line on fraud risk.

26 Excel: formulas, VLOOKUP, IF, errors

> **Q55:** Cells A1:A4 hold four plants' power output. Write a function in A5 for the average of the best three. / Explain the VLOOKUP formula. / Using IF, put the smallest of A1,B1,C1 into D1.
> ✎ **Asked in:** NESCO AM(ICT) 2018 · BTCL JAM 2024 · PGCB SAE 2021

```
A5:  =AVERAGE(LARGE(A1:A4,{1,2,3}))
D1:  =MIN(A1,B1,C1)        (or nested IF)
VLOOKUP: =VLOOKUP(lookup_value, table_range, col_index, FALSE)
```

=AVERAGE(LARGE(A1:A4,{1,2,3})) returns the mean of the three largest of the four cells (the exact exam answer). VLOOKUP finds a value in the first column of a range and returns a value from the given column number; FALSE = exact match.

MCQ as asked	answer	asked in
In Excel, every formula starts with —	**= (equals sign)**	Combined Bank Officer(IT) 2026
=Min(Average(B1,B2), Product(B3:B4)), B1=20 B2=24 B3=15 B4=20	Average=22, Product=300 → **Min = 22**	BB AD(General) 2023
#NAME? error occurs when —	**Function/name is misspelled** (ক); #REF! = bad reference	BB AD(General) 2023
User-developed instructions for Excel are called —	**Macro**	BB AD(General) 2021
Center-align text shortcut in MS Word	**Ctrl + E**	BB AD(General) 2023

Common Excel functions: `SUM`, `AVERAGE`, `COUNT`, `COUNTIF`, `MAX/MIN`, `LARGE/SMALL`, `IF`, `AND/OR`, `VLOOKUP/HLOOKUP`, `INDEX+MATCH`, `CONCATENATE`, `LEFT/RIGHT/MID`, `ROUND`, `NOW/TODAY`. **Reference types:** relative `A1`, absolute `$A$1` (fixed when copied), mixed `$A1`. **Errors:** `#NAME?` bad name/function · `#REF!` deleted reference · `#DIV/0!` divide by zero · `#VALUE!` wrong type · `####` column too narrow.

VLOOKUP pitfalls: the lookup column must be the *first* column of the range; use `FALSE` for an exact match (`TRUE` needs a sorted table); `col_index` counts from the range's first column, not the sheet's. For a left-side lookup use `INDEX+MATCH`. **Office shortcuts:** Ctrl+C/V copy/paste · Ctrl+Z undo · Ctrl+E center (Word) · F12 Save As · Ctrl+S save.

Exam trap: `=Product(B3:B4)` multiplies (15×20=300); `Average(B1,B2)` = 22; `Min` picks the smaller → 22, not 300. Read the nesting from the inside out.

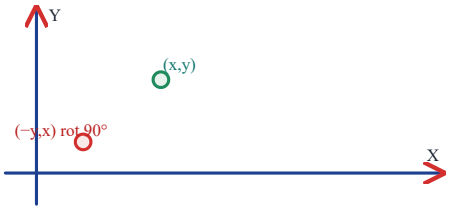
27 Transformations: reflection, rotation, shear

These appeared in the BUET M.Sc. admission test (counted as a job exam). 2D/3D transformation rules:

Transform	rule (2D/3D)
Reflection on XZ plane	(x, y, z) → (x, **−y**, z)
Rotation 90° about Z	(x, y) → **(−y, x)**
Shear by k on X axis	x′ = x + k·y, y′ = y

MCQ as asked	answer	asked in
A(3,4,1) reflected on the XZ plane →	**(3, −4, 1)**	BUET M.Sc. April 2021
Point (1,2,3): rotate 90° about Z axis →	**(−2, 1, 3)**	BUET M.Sc. April 2021
Triangle (1,1),(0,0),(1,0), shear k=2 on X →	**(3,1),(0,0),(1,0)**	BUET M.Sc. April 2021
Plane equidistant from P1(0,0,0) & P2(2,2,0)	**x + y − 2 = 0**	BUET M.Sc. April 2021
Orthographic projection: face is ___ to the picture plane	**Parallel**	BUET M.Sc. April 2021

Worked — rotate (1,2) 90° about Z: (x,y)→(−y,x) = (−2,1); z unchanged → **(−2,1,3)**. **Shear check:** x′=x+2y → (1+2·1, 1)=(3,1); (0,0)→(0,0); (1,0)→(1,0).



↑ rotate 90° about Z: (x, y) → (−y, x); reflection on XZ flips y only

Exam trap: Reflection on the **XZ plane** flips the **y** sign only (not x). Reflection on XY flips z; on YZ flips x. Match the missing axis to the flipped coordinate.

28 Keyboard, Office & device MCQs

MCQ as asked	answer	asked in
Shift, Ctrl and Alt keys are called —	**Modifier keys**	BB AD(General) 2026
Which is a modifier key of the keyboard?	**Ctrl / Shift / Alt**	Combined Bank Officer(IT) 2020
Which operation does F1 perform in most apps?	**Help**	Combined Bank Officer(IT) 2020
What does F3 stand for in most computers?	**Search / Find next**	BB AD(General) 2021
Key to show the Save-As box in MS Word	**F12**	BB AD(General) 2021
Which is an extension of a video file format?	**.avi / .mov / .mp4**	BB AD(General) 2021
Which is NOT a video file format? AVI/MOV/**JPEG**/JPG	**JPEG / JPG** (image)	BPSC AP 2022
Which is used as an image-file extension? .docx/.xls/**.jpg**	**.jpg**	BTCL JAM 2022
Which technology is used in a Compact Disk (CD)?	**Laser**	BPSC AP 2022
The job of ___ is to translate an array of dots into text	**OCR software**	Combined Bank Officer(IT) 2020
In a plasma display, gas is electrified by a grid of —	**Electrodes**	Combined Bank Officer(IT) 2020
In a spreadsheet, ___ helps you make sense of a worksheet's contents	**Chart / graph**	Combined Bank Officer(IT) 2020
OTG cable is NOT related to —	**Printing** (OTG links phone ↔ USB device)	BB AD(General) 2018
A device that is both input and output	**Touch screen / modem / network card**	BB AD(General) 2018
You can include a picture/object by using —	**Insert menu**	BB AD(General) 2018
Mixed tick-mark set: fastest memory? / which SQL runs last? / HTTP error port?	**Register (fastest) · ORDER BY/COMMIT last · port 80 (HTTP)**	BAERA Engineer(CSE) January 2021
Programs that help a computer manage its tasks/devices are called ___	**System software**	BPSC AME 2019
Which is a utility software? Windows / Oracle / **Antivirus** / MS Word	**Antivirus**	BPSC AME 2017
First Bangla software for the visually impaired (আইলিশ / আইসাইট / **আইডট** / আইলাইট)	**আইডট (i-dot)**	17th NTRCA School Prelim (2022)

Exam trap: **JPEG/JPG, PNG, GIF, BMP** = image; **AVI, MOV, MP4, MKV, WMV** = video; **MP3, WAV** = audio; **DOCX, XLSX, PDF** = document. Sort the odd one out by media type.

29 Internet, storage & misc MCQs

MCQ as asked	answer	asked in
Which is NOT a web browser? Firefox/**Facebook**/Chrome/Safari	**Facebook**	BPSC AP 2022 · BTCL JAM 2022
In which year did Gmail start? 1998/1988/**2004**/2021	**2004**	BCC AP 2021
Which is a correct email format?	**name@domain.com**	BCC AP 2021
Which language helps you learn Android programming?	**Java** (also Kotlin)	BCC AP 2021
Full form of DVD	**Digital Versatile Disc**	BB AD(General) 2018
When browsing, the browser stores some data as a —	**Cookie**	DESCO AE(CSE) 2016
Which is NOT a true statement about the Recycle Bin?	**"Disk space increases by sending files to it"** (false)	Combined Bank AHE/AE(IT) 2021
.NET can be used for —	**All of the above** (desktop, microservices/containers, cloud)	Combined Bank SO(IT) 2021
Java Virtual Machine —	**Manages memory & gives Java a portable execution environment**	Combined Bank SO(IT) 2021
Main task for a computer in mapping geographical data	**Data retrieving and drawing**	BPSC AP 2022
Why is RAM unsuitable as permanent storage?	**Volatility**	BTRC AD(Technical) 2019 · BTRC SAD 2021

Q56: Write short notes on DNS, cloud server and firewall. / Business Intelligence model. / Decision tree for playing football under weather conditions. / Ergonomics: how to judge a good chair. / SQL query to retrieve the uncommon names from two tables.

✎ **Asked in:** Intl Crimes Tribunal Hardware Technician 2024 · Combined Bank SO(IT) 2023 · BAERA Engineer(CSE) 2023 · Combined Bank SO(IT) 2020 · RAKUB ADBA 2020

DNS maps names→IP; **cloud server** = a virtual server rented over the internet; **firewall** filters traffic by rules. **BI** turns data into dashboards/reports for decisions (ETL → data warehouse → OLAP → visualisation). **Decision tree**: branch on weather (outlook/humidity/wind) to a Play/Don't-Play leaf. **Ergonomics** (good chair): adjustable height, lumbar support, armrests, breathable, feet flat, screen at eye level. **Uncommon names** (symmetric difference) = names in exactly one table (SQL: `FULL JOIN … WHERE a IS NULL OR b IS NULL`, or two `EXCEPT` queries UNIONed).

★ Night before the exam

1. **System vs application software** (★ most asked): system runs the machine (OS, compiler, driver, utility); application does a user task (Word, browser). Antivirus = utility = system software.
2. **Generations**: tube → transistor → IC → **microprocessor (4th)** → AI. Each = smaller, faster, cheaper.

3. **CPU** = CU + ALU + registers; **buses** = data (2-way) / address (1-way) / control. IPv6 = 128 bits.
4. **OCR** characters · **OMR** marks · **MICR** magnetic cheque digits · **RFID** radio, no line-of-sight.
5. **RAM** volatile, read/write · **ROM** permanent, holds BIOS · **SSD** no moving parts, fast.
6. 1 byte = 8 bits; 1 KB = 1024 B (binary) / 1000 (SI). Unicode char = 16 bits. MB = storage, Mbps = speed.
7. **BIOS** firmware in ROM → POST → bootloader → OS; settings in CMOS (battery).
8. **Cloud**: on-demand resources over the internet. **IaaS** rent infra · **PaaS** deploy code only · **SaaS** ready app. Deploy-only startup ⇒ PaaS.
9. **VM** = own guest OS on a hypervisor (Type 1 bare-metal / Type 2 hosted); **container** shares the host kernel (Docker), lighter. VM count = min over CPU/RAM/storage.
10. **Data centre**: cooling is the biggest energy cost; PUE → 1 is ideal. Preventive (before) vs corrective (after) maintenance. DR = RTO + RPO + backups.
11. **Scaling**: vertical = bigger box; horizontal = more boxes + load balancer; auto-scaling adds/removes nodes.
12. **HTML** element = tags + content; HTML5 semantic: header/footer/section/article. **CSS** box: content→padding→border→margin; 3 types inline/internal/external.
13. **GET** in URL, cacheable, read · **POST** in body, for login/submit. **HTTP**: 2xx ok, 4xx client, 5xx server (500 = internal error).
14. **SOAP** = XML only; **REST** = lightweight over HTTP verbs. **XSS** fix: escape output, CSP, HttpOnly cookies.
15. **Blockchain** = block (data + hash + previous hash), immutable chain. **IR 4.0** = AI + IoT + Big Data + cloud + robotics.
16. **E-commerce**: B2C, B2B, C2C, G2C. **EPS** flow: customer → gateway → acquirer → network → issuer. **WWW** ⊂ internet.

✓ Practice MCQs (all asked; answers at the bottom)

1. Which is NOT a layer of cloud computing? (a) CaaS (b) IaaS (c) PaaS (d) SaaS ✎ Combined Bank AHE/AE(IT) 2021

2. BIOS stands for: (a) Basic Input/Output System (b) Basic Interrupt/Output System (c) Binary I/O Set (d) None ✎ BTCL JAM 2022

3. A Unicode character uses: (a) 8 bits (b) 16 bits (c) 20 bits (d) 24 bits ✎ BCC AP 2021

4. HTTP status code 500 means: (a) Bad Request (b) Unauthorized (c) Internal Server Error (d) Not Found ✎ BB AD(ICT) 2025

5. Microprocessor was first used in which generation? (a) 2nd (b) 3rd (c) 4th (d) 5th ✎ DESCO AE(CSE) 2016

6. Which is NOT a web browser? (a) Firefox (b) Facebook (c) Chrome (d) Safari ✎ BPSC AP 2022

7. Which tag writes JavaScript in HTML? (a) <js> (b) <script> (c) <style> (d) <code> ✎ BCC AP 2023

8. The CSS area outside the border is: (a) padding (b) content (c) margin (d) outline ✎ 15th NTRCA ICT

9. Which is main memory? (a) HDD (b) ROM (c) RAM (d) Floppy ✎ DoICT AP 2020

10. In Excel, every formula starts with: (a) + (b) = (c) @ (d) # ✎ Combined Bank Officer(IT) 2026

Answers: 1 a · 2 a · 3 b · 4 c · 5 c · 6 b · 7 b · 8 c · 9 c · 10 b

✎ Written practice (model answer marked)

Q: What is cloud computing? Explain IaaS, PaaS and SaaS. (10)

✎ **Asked in:** Combined Bank AP(SB & Janata) 2026 · DMTCL AE(ICT) 2023 · BB AME 2023

Cloud computing = on-demand delivery of computing resources over the internet, pay-as-you-go (2).

IaaS = rent VMs/storage/network, you manage OS & app — AWS EC2 (2).

PaaS = platform + runtime, you deploy only your code — Heroku, App Engine (2).

SaaS = ready software, just use it — Gmail, Office 365 (2).

Add a diagram (who-manages-what stack) + benefits/one example each (2).

Memory trick: ICS answers: **definition → table/diagram → BD example → one trap line**. A cloud/VM answer always needs the who-manages-what stack or the hypervisor diagram.

Before the exam ✓: service models (IaaS/PaaS/SaaS), VM vs container, HTTP 2xx/4xx/5xx, CSS box model, OCR/OMR/MICR, generations, RAM vs ROM, byte units.